Joseph-Vu Trinh

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Education

Virginia Tech Blacksburg, VA
Bachelor of Science - Computer Science, Minor in Artificial Intelligence *Graduating May 2028*
Relevant Coursework: Data Structures & Algorithms, Object Oriented Programming, Computer Systems, Machine Learning

Experience

Web Developer July 2026 – Present
SASE at Virginia Tech *Blacksburg, VA*
- Rebuilt the club site in **Next.js**, **TypeScript**, and **Tailwind CSS**, replacing a legacy site with a mobile-first architecture that increased site visits by **53%**.
- Working with organization leadership to turn requirements into designs, maintaining the live site with event listings and image updates each semester.

Software Developer June 2026 – Present
Virginia Tech Department of Computer Science *Blacksburg, VA*
- Shipped backend REST API services in **Python** and **Flask** for DBWorkout, a SQL practice platform deployed to **170** students in Virginia Tech's CS4604 database course, where **90%** reported improved learning value post-deployment.
- Executed student queries against isolated **PostgreSQL** sandboxes via SQLAlchemy, diffing result sets against expected output to grade submissions automatically instead of by hand.
- Added **OpenAI API** hint generation from a student's failed query and exercise schema, driving a feedback overhaul after only **42%** of surveyed students rated the original feedback actionable.

Undergraduate Researcher Jan 2026 – June 2026
Virginia Tech Department of Computer Science *Blacksburg, VA*
- Engineered a machine learning pipeline in **Python** using **OpenPose** to extract skeletal pose and gaze features from **180** one-minute classroom video segments across **300+** students.
- Classified student engagement from the extracted features without storing identifiable imagery, improving classification accuracy by **25%** over the prior approach.
- Benchmarked LLM-generated engagement observations against human annotations using the **OpenAI API**, reducing malformed model responses by **85%**.

Full Stack Developer Intern Sep 2024 – Dec 2024
Engauge *College Park, MD*
- Built a collaborative whiteboard in **React** and **TypeScript** with **Firestore** persistence, synchronizing state across **30+** concurrent clients and contributing to a **30%** increase in student engagement during live sessions.
- Modeled Firestore documents to scope whiteboards to individual students while giving instructors real-time visibility across all submissions in a session.
- Rebuilt the front-end component architecture of a classroom web application, cutting re-renders for sessions of **150+** students.
- Implemented a live transcription and chat feature, wiring **WebSocket** updates and **GCP Cloud Functions** to persist transcripts and conversation history.

Projects

Professor Search Platform | *VTHacks* | *Python, React, Databricks Agent Bricks, GraphQL*
- Scraped and ingested **16,000** records spanning **577** professors and **3,267** reviews through asynchronous Reddit collection and paginated **GraphQL**, with filtering, deduplication, retry logic, and rate-limit handling.
- Integrated the curated dataset with **Databricks Agent Bricks** for an LLM-powered recommendation system answering natural-language questions about professors and courses, and developed the supporting React interface.

Playlist Curator | *React, TypeScript, FastAPI, Anthropic API, Spotify Web API, librosa*
- Developed a playlist curation tool in **React**, **TypeScript**, and **FastAPI** that scores Spotify tracks against user-defined themes for review before applying changes.
- Analyzed playlists of arbitrary size through **10+** backend endpoints and local audio signal analysis with librosa, combining acoustic features with Spotify metadata.

Technical Skills

Languages: Python, Java, C/C++, C#, SQL, JavaScript, TypeScript, Swift
Frameworks: Flask, FastAPI, React, Next.js, Node.js, Tailwind CSS, React Native, SwiftUI
Tools & Technologies: Git, Linux/Unix, Claude Code, Codex, Figma, PostgreSQL, Docker, CI/CD, REST APIs, WebSockets, AWS, GCP, Firebase/Firestore, MongoDB (NoSQL), JUnit